

Course Code: CS1385		
Name of the Programme: BCA (Hons.)		
Name of the Course: Web and UX Design		
Semester: I		
Course Credit	No. of Hours Per Week (L + T + P)	Total No. of Teaching Hours
3	1:1:2	60
Course Objectives:		
a) Introduce the core technologies and principles of modern web development.		
b) Develop the ability to design and create user-centered, accessible websites.		
c) Familiarize students with the UX design process, including planning, prototyping, and usability testing.		
d) Provide hands-on experience in implementing and testing web interfaces, laying a foundation for further learning and professional growth.		

Course outline (Syllabus of the course)

Module - 1: Web and HTML Fundamentals	No. of Hours
	3 Hrs.
Introduction to the Web and Web Development, Basic Structure of an HTML Document, Basic HTML Tag and Attributes, Creating Links, Lists, Tables, and Forms	
Module - 2 : CSS Fundamentals	No. of Hours
	3 Hrs.
CSS Selectors, Styling Text, Boxes, and Layouts, Introduction to Responsive Design.	
Module - 3 : JavaScript Basics	No. of Hours
	3 Hrs.
JavaScript Syntax and Programming Concepts, Manipulating the DOM (Document Object Model).	
Module - 4 : UX Design Fundamentals	No. of Hours
	3 Hrs.
Introduction to UX Design and Its Importance, User Research and Personas, Wireframing and Prototyping, Usability Testing and Basic Accessibility Standards.	

Module - 5 : Integrated Project	No. of Hours
	3 Hrs.
Overview of Project Requirements and Planning, Integration of HTML, CSS, JavaScript, and UX Design Principles and Project Management.	

Course Outcomes: After completing the course, the students will be able to: -	
a)	Understand the basic structure of HTML and apply CSS to design well-structured and visually styled web pages.
b)	Analyze and manipulate the Document Object Model (DOM) using JavaScript to create dynamic and interactive web pages.
c)	Understand the principles of user experience (UX) design to create user-friendly and accessible web interfaces.
d)	Apply web development and UX principles to develop a complete web project from concept to implementation.

Text Books	
1	Duckett, J. (2011). <i>HTML and CSS: Design and build websites</i> (1st ed.). John Wiley & Sons. ISBN: 978-1-118-00818-8
2	Duckett, J. (2014). <i>JavaScript and jQuery: Interactive front-end web development</i> . John Wiley & Sons. ISBN: 978-1-118-53164-8 / 978-1-118-87165-2
3	Unger, R., & Chandler, C. (2012). <i>A project guide to UX design: For user experience designers in the field or in the making</i> (2nd ed.). New Riders. ISBN: 978-0-321-81538-5

Reference Books	
1	Robbins, J. N. (2018). <i>Learning web design: A beginner's guide to HTML, CSS, JavaScript, and web graphics</i> . Shroff/O'Reilly. ISBN: 978-1-491-46020-2
2	Haverbeke, M. (2011). <i>Eloquent JavaScript: A modern introduction to programming</i> . No Starch Press. ISBN: 978-1-59327-282-1
3	Norman, D. A. (2013). <i>The design of everyday things</i> (2nd ed.). Basic Books. ISBN: 978-0-465-05065-9
4	Gothelf, J., & Seiden, J. (2016). <i>Lean UX: Designing great products with agile teams</i> . O'Reilly Media, Inc. ISBN: 978-1-491-95360-0
5	Krug, S. (2014). <i>Don't make me think, revisited: A common sense approach to web usability</i> . New Riders. ISBN: 978-0-321-96551-6

Tutorial Components**15 Hrs**

1	Introduction to Web Development: Learn how the web works, the roles of browsers and servers, and create your first HTML webpage.
2	HTML Structure and Basic Tags: Understand HTML elements, attributes, and semantic tags to structure webpages effectively.
3	Organizing Content: Use tables, lists, and forms to organize and collect data on web pages.
4	Introduction to CSS: Explore CSS syntax, selectors, and ways to style webpages using inline, internal, and external methods.
5	Styling Text and Colors: Apply fonts, text styles, colors, and background designs to enhance webpage aesthetics.
6	CSS Layout Principles: Master the box model, positioning, and flexbox to create structured webpage layouts.
7	Responsive Design: Use media queries and fluid layouts to make webpages mobile-friendly.
8	JavaScript Syntax Basics: Learn variables, operators, and control structures to write simple JavaScript code.
9	DOM Manipulation: Modify HTML content dynamically using the Document Object Model.
10	Event Handling: Add interactivity with event listeners like clicks, keypresses, and form validations.
11	Introduction to UX Design: Understand user-centered design and the importance of creating user-friendly interfaces.
12	User Research: Conduct surveys and interviews to gather insights for designing better webpages.
13	Wireframing and Prototyping: Plan layouts and navigation using tools to create low-fidelity wireframes.
14	Accessibility in Web Design: Follow WCAG standards to make webpages inclusive for all users.
15	Integrated Project: Combine HTML, CSS, JavaScript, and UX principles to build a complete, functional, and visually appealing webpage.

Laboratory Components**30 Hrs**

1	Write a program to develop Basic Webpage with HTML.
2	Write a program to develop creating a personal profile page.
3	Write a program to design a feedback form using various HTML form elements.
4	Write a program to build a table to display a schedule.
5	Write a program to style a basic webpage with different font styles, colors, and background images.
6	Write a program to create a two-column layout using floats and a grid system.
7	Write a program to make a web page responsive using media queries.
8	Write a program to style a form and table for better usability and appearance.
9	Write a JavaScript program to perform basic arithmetic operations.

10	Write a JavaScript program to Manipulate the content of HTML elements.
11	Write a program to create an interactive form that validates user input.
12	Build a simple to-do list application with dynamic DOM manipulation
13	Write a program to create simple interactive web pages using HTML, CSS, JavaScript.
14	Write a program to design a sitemap and navigation structure using HTML, CSS.
15	Write a program to create a basic webpage with a heading, paragraph, and image.
16	Write a program to create a profile page with biography, contact information, and social media links using HTML and CSS. Note: All pages should have consistent styling.
17	Write a program to design a feedback form using various form elements with HTML and CSS.